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Thermodynamic assessment of the Al–Cu–Er system

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ABSTRACT

The Cu–Er binary system had been thermodynamically assessed with the CALPHAD approach. The solution phases including Liquid, Fcc and Hcp were treated as substitutional solution phases, of which the excess Gibbs energies were formulated with the Redlich–Kister polynomial function. All the binary intermetallic compounds were treated as stoichiometric phases. Combining with the thermodynamic parameters of the Al–Cu and Al–Er binary systems cited from the literature, the Al–Cu–Er ternary system was thermodynamically assessed. The calculated phase equilibria were in good agreement with the experimental data.

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1. Introduction

Rare earth elements have been used in aluminum alloys for many years to improve the tensile strength, heat resistance and corrosion resistance, etc. [1]. For example, when Sc is added into Al alloys, a thermo-stable $L1_2$-type ($AuCu_3$) Al_3Sc phase will form in the Al solid solution, which could inhibit the recrystallization and the grain growth, and significantly improve the high temperature properties of the alloys [2].

Similar to the case for the element Sc, when Er was added into Al alloys an L12-type ($AuCu_3$) phase Al_3Er also formed. This process could significantly refine grains and improve the mechanical properties of the alloy. Additionally, it could accelerate the aging process and improve the effect of aging strengthening [3–6]. Zhao [7] studied the Al–Zn–Mg–Cu alloy with a small addition of Er and found that the Er mainly existed in the Al_3Er phase. The dispersed Al_3Er phase acted as the site for heterogeneous nucleation of $\eta(MgZn_2)$ phase and enhanced the mechanical properties. However, Li [8] found that the Er added into Al–Cu–Mg–Ag–Zr alloy mainly segregated at the grain boundary during solidification and resulted in a ternary Al_8Cu_4Er phase. It could also decrease the distance of interdendritic arms.

Alloy phase diagrams were useful guides in the development of new materials, as well as in the improvement of the performance of existing materials. The purpose of the present work was to evaluate the Al–Cu–Er ternary system and develop a thermodynamic description of the system by means of the CALculation of the PHAse Diagram (CALPHAD) technique [9].

2. Experiment information

2.1. The Cu–Er binary system

Love [10] and Kato and Copeland [11] investigated the Er-rich region of the Cu–Er binary system. Both investigations reported that the solubility of Cu in Hcp_Er phase was less than 0.53 at.%, as well as a eutectic reaction between CuEr and Hcp_Er. However, the differences between these two researches rested on the temperature and liquid composition of the eutectic reactions. According to Love [10], the eutectic temperature was found to be 860 °C and the composition was 75.5 at.% Er, while these were given as 845 °C and 71.8 at.% Er by Kato and Copeland [11].

Buschow [12] identified three compounds: Cu_5Er (peritectic formation), Cu_2Er (congruent formation) and CuEr (congruent formation) by means of X-ray diffraction, differential thermal analysis (DTA) and metallography in this system. Moreover, his work also indicated the existence of two additional phases containing about 20 at.% Er, which were designated as Cu_xEr and Cu_yEr, respectively, without exact stoichiometries. His investigation indicated the occurrence of four eutectic reactions: Cu–Cu_5Er (895 °C, 9.5 at.% Er); Cu_xEr–Cu_2Er (885 °C, 30 at.% Er); Cu_2Er–CuEr (905 °C, 40 at.% Er); and CuEr–Hcp_Er (880 °C, 69.5 at.% Er). It was also found that the solid solubility of Cu in Er could be negligible. Similarly, there was no evidence of terminal solid solubility of Er in Cu.

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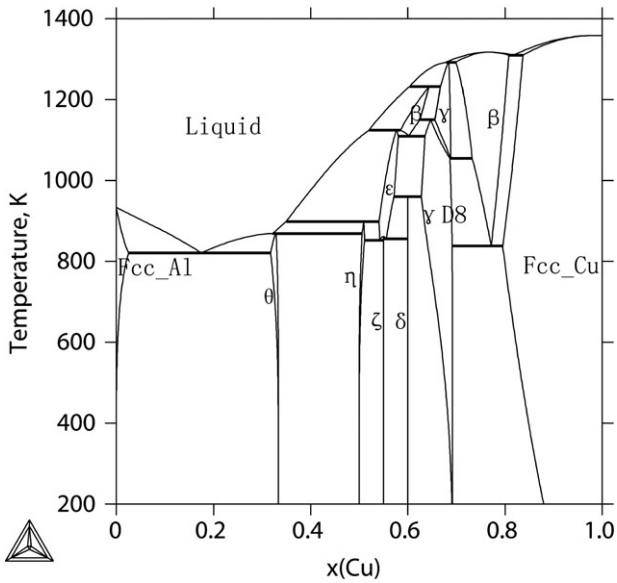

Fig. 1. The Al–Cu phase diagram calculated by Witusiewicz et al. [19].

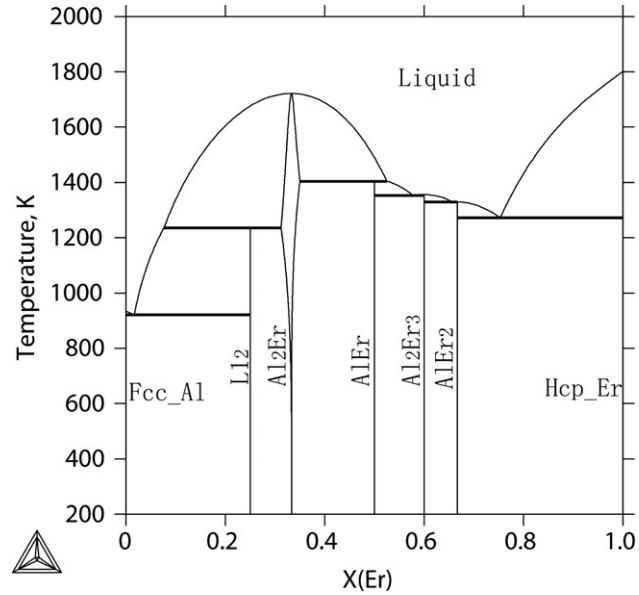

Fig. 2. The Al–Er phase diagram calculated by Cacciamani et al. [20].

Subramanian and Laughlin [13] accepted most of the aforementioned phase diagram information when they assessed this binary system. Moreover, according to the phase diagrams of Cu–Dy [14], Cu–Gd [15] and Cu–Ho [16] systems, Subramanian and Laughlin [13] suggested Cu_xEr and Cu_yEr to be Cu_9Er_2 and Cu_7Er_2 phases.

We have reassessed this system in this work. Similarly to the case for Subramanian and Laughlin [13], most of the above mentioned phase equilibria information was accepted in the thermodynamic optimization process except the stable temperature range of the Cu_7Er_2 phase. From the result of Subramanian and Laughlin [13], the Cu_7Er_2 phase was stable over the whole temperature range. But in Cu–Dy, Cu–Gd and Cu–Ho systems, the Cu_7RE_2 phases existed only within a limited temperature range. In this work, the Cu_7Er_2 phase was taken to be stable only at the elevated temperature.

As for the thermodynamic properties, only heats of mixing of liquid Cu-rich Cu–Er alloys were measured using calorimetry by Nikolaenko et al. [17]. Besides this, no other thermodynamic property of this system was reported.

2.2. The Al–Cu and Al–Er binary systems

The Al–Cu system was thermodynamically studied by Saunders [18]. Recently, Witusiewicz et al. [19] reassessed the liquid and $\gamma D8_3$ phases of the Al–Cu phase diagram on the basis of new experimental thermodynamic data on the $\gamma D8_3$ phase. In this work, the thermodynamic parameters by Witusiewicz et al. [19] were adopted as in Fig. 1.

The only available description of the Al–Er system was performed by Cacciamani et al. [20]. Fig. 2 is the calculated phase diagram based on their parameters.

2.3. The Al–Cu–Er ternary system

The only available experimental result on the phase diagram of the ternary Al–Cu–Er system was the 600 °C isothermal section of Kuz'ma et al. [21]. Subsequently, the experimental phase diagram had been assessed by Riani et al. [22]. It was reported that there were seven ternary compounds: τ_1 to τ_7, shown in Fig. 3. In this ternary system, τ_1-$ErCu_4Al_8$, τ_4-Al_3CuEr, τ_5-$Al_5Cu_3Er_2$, τ_6-$Al_9Cu_6Er_5$ and τ_7-AlCuEr were stoichiometric compounds; meanwhile, Al and Cu could substitute each other partially in two

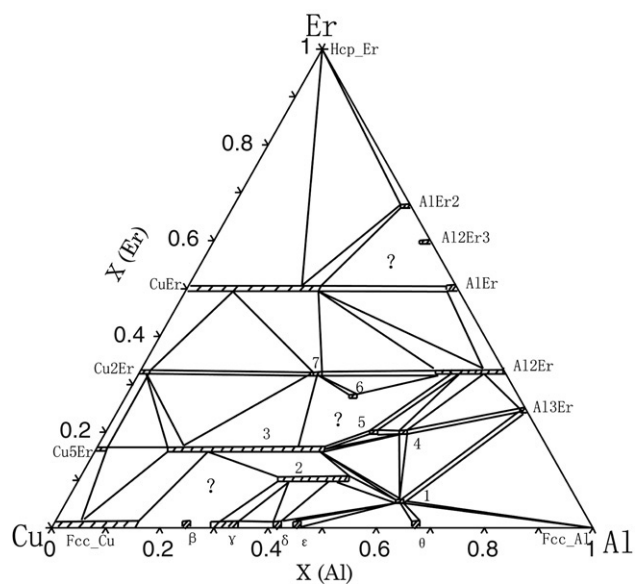

Fig. 3. The 600 °C experimental phase diagram of Al–Cu–Er system of Kuz'ma et al. [21].

semi-stoichiometric phases, τ_2-$(Al, Cu)_{17}Er_2$ and τ_3-$(Al, Cu)_5Er$. The structures of these phases were listed in Table 1. At 600 °C, Cu_2Er dissolved up to 3 at.% Al, CuEr up to 25 at.% Al, and Al_2Er up to 13 at.% Cu.

The experimental 600 °C isothermal section of this ternary system was inconsistent with the boundary binary systems which were adopted in this work. At the Al–Cu edge of this ternary system, Kuz'ma et al. [21] found the θ-Al_2Cu phase existed at 600 °C; however, according to Saunders [18] this phase should not be stable at this temperature. Furthermore, with the result of Saunders [18], the liquid phase and η phase should be stable at 600 °C, but in the research of Kuz'ma et al. [21], these two phases were not found. In order to have accordance with the binary Al–Cu system, the liquid and η phases were treated as stable at 600 °C in the present optimization but the θ-Al_2Cu phase was not. Additionally, at the Cu–Er edge of this ternary system, Kuz'ma et al. [21] just reported three compounds (CuEr, Cu_2Er and Cu_5Er phases). But from our calculated result, the Cu_9Er_2 phase should

Table 1
Crystallographic data for intermetallic phases

Phase	Lattice parameters			Structure type, Pearson symbol, space group
	a (nm)	b (nm)	c (nm)	
θ	0.6067		0.4877	$CuAl_2$, $tI12$, $I4/mcm$
η	1.2066	0.4105	0.6913	CuAl, $mC20$, $C2/m$
ζ	0.40972	0.71313	0.99793	$Cu_{11.5}Al_9$, oI24, $Imm2$
ε	0.4146		0.5063	Ni_2In, $hP6$, $P63/mmc$
γD83	0.87023			Cu_9Al_4, $cP52$, $P\bar{4}3m$
γ				Cu_5Zn_8, $cI52$, $I\bar{4}3m$
β	0.2946			W, $cI2$, $Im\bar{3}m$
$L1_2$	0.4214			$AuCu_3$, $cP4$, $Pm\bar{3}m$
Al_2Er	0.77967			$MgCu_2$, $cF24$, $Fd\bar{3}m$
AlEr	0.5570	0.5801	1.1272	AlEr, $oP16$, $Pmma$
Al_2Er_3	0.8123			Al_2Zr_3, $tP20$, $P42/mnm$
$AlEr_2$	0.6516	0.5015	0.9279	Co_2Si, $oP12$, $Pnma$
Cu_5Er	0.7003			$AuBe_5$, $cF24$, $F\bar{4}3m$
Cu_9Er_2				?
Cu_7Er_2				?
Cu_2Er	0.4275	0.6726	0.7265	$CeCu_2$, $oI12$, $Imma$
CuEr	0.3425			CsCl, $cP2$, $Pm\bar{3}m$
τ_1-Al_8Cu_4Er	0.8712		0.5130	$Mn_{12}Th$, $tI26$, $I4/mmm$
τ_2-$(Al, Cu)_{17}Er_2$	0.8630		0.5029	Th_2Zn_{17}, $hR19$, $R\bar{3}m$
τ_3-$(Al, Cu)_5Er$	0.5029		0.4139	$CaCu_5$, $hP6$, $P6/mmm$
τ_4-Al_3CuEr	0.4184	0.4112	0.9773	Al_3CuHo, $oI10$, $Immm$
τ_5-$Al_5Cu_3Er_2$				?
τ_6-$Al_9Cu_6Er_5$				?
τ_7-AlCuEr	0.6975		0.4003	Fe_2P, $hP9$, $P\bar{6}2m$

exist at this temperature. In this study the Cu_9Er_2 phase was thought to be stable at 600 °C.

As there are no available experimental or calculated thermochemistry data, the Al–Cu–Er system is assessed only on the basis of the ternary phase diagram information.

3. Thermodynamic model

3.1. Solution phases

The lattice stabilities for element Al, Cu and Er are obtained from Dinsdale [23]. An ordinary substitutional solution model is employed to describe Liquid, Fcc and Hcp terminal solutions. The mole Gibbs energy of a solution phase Φ (Φ = Liquid, Fcc, Hcp) can be represented as a sum of the weighted Gibbs energy for the pure components, the ideal entropy term describing a random mixing of the components, and the excess Gibbs energy describing the degree of deviation from ideal mixing, i.e.,

$$G^{\phi} = \sum_{i=\mathrm{Al,Cu,Er}} x_i\, {}^0G_i^{\phi} + RT \sum_{i=\mathrm{Al,Cu,Er}} x_i \ln(x_i) + {}^{ex}G^{\phi} \tag{1}$$

where

$$\begin{aligned} {}^{ex}G^{\phi} = {} & x_{\mathrm{Al}}x_{\mathrm{Cu}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Cu}})^j\, {}^{(j)}L^{\phi}_{\mathrm{Al,Cu}} \\ & + x_{\mathrm{Al}}x_{\mathrm{Er}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Er}})^j\, {}^{(j)}L^{\phi}_{\mathrm{Al,Er}} \\ & + x_{\mathrm{Cu}}x_{\mathrm{Er}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Cu}} - x_{\mathrm{Er}})^j\, {}^{(j)}L^{\phi}_{\mathrm{Cu,Er}} + x_{\mathrm{Al}}x_{\mathrm{Cu}}x_{\mathrm{Er}}L^{\phi}_{\mathrm{Al,Cu,Er}} \end{aligned} \tag{2}$$

where ϕ denotes the solution phases, x_i (i = Al, Cu and Er) denotes the mole fraction of component i, and ${}^0G_i^{\phi}$ is the molar Gibbs energy of pure element i in the structural state of ϕ.

${}^{(j)}L^{\phi}_{\mathrm{Al,Cu}}$ is taken from Witusiewicz et al. [19] and ${}^{(j)}L^{\phi}_{\mathrm{Al,Er}}$ is taken from Cacciamani et al. [20]. ${}^{(j)}L^{\phi}_{\mathrm{Cu,Er}}$ is expressed as

$${}^{(j)}L^{\phi}_{\mathrm{Cu,Er}} = A_j + B_jT \tag{3}$$

where the A_j and B_j are constants to be optimized.

$L^{\phi}_{\mathrm{Al,Cu,Er}}$ are ternary interaction parameters. Due to the lack of experimental data, they are set to zero.

3.2. Binary intermetallic phases

Most of the binary intermetallic phases are treated as stoichiometric phases, which are described as M_pN_q According to Neumann–Kopp rule, the Gibbs energy of the phase M_pN_q is formulated as

$$G_{MpNq} = \frac{p}{p+q}\, {}^0G_M^{\mathrm{HSER}} + \frac{q}{p+q}\, {}^0G_N^{\mathrm{HSER}} + A + BT \tag{4}$$

where M and N denote Al, Cu, or Er, A and B are the adjusted parameters being optimized in the present work.

According to Kuz'ma et al. [21], the Al_2Er, CuEr and Cu_2Er phases have a homogeneity region in the Al–Cu–Er ternary system. The Gibbs energy expressions for the CuEr and Cu_2Er phases are formalized as

$$\begin{aligned} G^{(\mathrm{Al,Cu})_x\mathrm{Er}_y} = {} & Y^{\mathrm{I}}_{\mathrm{Al}}G_{\mathrm{Al:Er}} + Y^{\mathrm{I}}_{\mathrm{Cu}}G_{\mathrm{Cu:Er}} + \frac{x}{x+y}RT(Y^{\mathrm{I}}_{\mathrm{Al}} \ln Y^{\mathrm{I}}_{\mathrm{Al}} \\ & + Y^{\mathrm{I}}_{\mathrm{Cu}} \ln Y^{\mathrm{I}}_{\mathrm{Cu}}) + Y^{\mathrm{I}}_{\mathrm{Al}}Y^{\mathrm{I}}_{\mathrm{Cu}} \left(\sum_{j=0,1\ldots} {}^jL_{\mathrm{Al,Cu:Er}}(Y_{\mathrm{Al}} - Y_{\mathrm{Cu}})^j \right) \end{aligned} \tag{5}$$

where $G_{\mathrm{Cu:Er}}$ is taken from the binary system, and $G_{\mathrm{Al:Er}}$ is expressed as

$$G_{\mathrm{Al:Er}} = \frac{x}{x+y}\, {}^0G_{\mathrm{Al}}^{\mathrm{Fcc}} + \frac{y}{x+y}\, {}^0G_{\mathrm{Er}}^{\mathrm{Hcp}} + A + BT \tag{6}$$

which represents the Gibbs energy of the assumed AlEr compound with CuEr structure or the Al_2Er compound with Cu_2Er structure.

A and B are the adjusted parameters being optimized in the present work.

The Gibbs energy expression for the Al_2Er phase is formalized as

$$\begin{aligned} G^{(\mathrm{Al,Cu,Er})2(\mathrm{Al,Cu,Er})1} = {} & \sum_i \sum_j Y^{\mathrm{I}}_i Y^{\mathrm{II}}_j G_{i:j} + 2RT(Y^{\mathrm{I}}_{\mathrm{Al}} \ln Y^{\mathrm{I}}_{\mathrm{Al}} + Y^{\mathrm{I}}_{\mathrm{Cu}} \ln Y^{\mathrm{I}}_{\mathrm{Cu}} \\ & + Y^{\mathrm{I}}_{\mathrm{Er}} \ln Y^{\mathrm{I}}_{\mathrm{Er}}) + RT(Y^{\mathrm{II}}_{\mathrm{Al}} \ln Y^{\mathrm{II}}_{\mathrm{Al}} + Y^{\mathrm{II}}_{\mathrm{Cu}} \ln Y^{\mathrm{II}}_{\mathrm{Cu}} + Y^{\mathrm{II}}_{\mathrm{Er}} \ln Y^{\mathrm{II}}_{\mathrm{Er}}) \end{aligned}$$

$$+\sum_i\sum_j\sum_k Y_i^{\mathrm{I}}Y_j^{\mathrm{I}}Y_k^{\mathrm{II}}\sum_{v=0,1\ldots}{}^{v}L_{i,j:k}(Y_i^{\mathrm{I}}-Y_j^{\mathrm{I}})^v$$
$$+\sum_i\sum_j\sum_k Y_k^{\mathrm{I}}Y_i^{\mathrm{I}}Y_j^{\mathrm{II}}\sum_{v=0,1\ldots}{}^{v}L_{k:i,j}(Y_i^{\mathrm{II}}-Y_j^{\mathrm{II}})^v \quad (7)$$

where i, j, k denote Al, Cu, Er, while the $G_{\mathrm{Al:Er}}$, $G_{\mathrm{Al:Er}}$, $G_{\mathrm{Er:Al}}$, and $G_{\mathrm{Er:Er}}$ are taken from binary system, and other parameters are optimized in this work.

3.3. Ternary intermetallic compounds

In the ternary system, τ_1-$ErCu_4Al_8$, τ_4-Al_3CuEr, τ_5-$Al_5Cu_3Er_2$, τ_6-$Al_9Cu_6Er_5$ and τ_7-AlCuEr are modeled as stoichiometric phases ($Al_xCu_yEr_z$), and τ_2-$(Al, Cu)_{17}Er_2$ and τ_3-$(Al, Cu)_5Er$ are treated as semi-stoichiometric phases ($(Al, Cu)_xEr_y$). The Gibbs energy expression for any of stoichiometric compounds is written as

$$G_{\mathrm{Al}_x\mathrm{Cu}_y\mathrm{Er}_z}=\frac{x}{x+y+z}\,{}^0G_{\mathrm{Al}}^{\mathrm{Fcc}}+\frac{y}{x+y+z}\,{}^0G_{\mathrm{Cu}}^{\mathrm{Fcc}}+\frac{z}{x+y+z}\,{}^0G_{\mathrm{Er}}^{\mathrm{Hcp}}+A+BT. \quad (8)$$

And for any of the semi-stoichiometric compounds, this is written as

$$G^{(\mathrm{Al,Cu})_x\mathrm{Er}_y}=Y_{\mathrm{Al}}^{\mathrm{I}}G_{\mathrm{Al:Er}}+Y_{\mathrm{Cu}}^{\mathrm{I}}G_{\mathrm{Cu:Er}}+\frac{x}{x+y}RT(Y_{\mathrm{Al}}^{\mathrm{I}}\ln Y_{\mathrm{Al}}^{\mathrm{I}}+Y_{\mathrm{Cu}}^{\mathrm{I}}\ln Y_{\mathrm{Cu}}^{\mathrm{I}})+Y_{\mathrm{Al}}^{\mathrm{I}}Y_{\mathrm{Cu}}^{\mathrm{I}}\left(\sum_{j=0,1\ldots}{}^{j}L_{\mathrm{Al,Cu:Er}}(Y_{\mathrm{Al}}-Y_{\mathrm{Cu}})^j\right) \quad (9)$$

where

$$G_{\mathrm{Al:Er}}=\frac{x}{x+y}\,{}^0G_{\mathrm{Al}}^{\mathrm{Fcc}}+\frac{y}{x+y}\,{}^0G_{\mathrm{Er}}^{\mathrm{Hcp}}+A+BT \quad (10)$$

$$G_{\mathrm{Cu:Er}}=\frac{x}{x+y}\,{}^0G_{\mathrm{Cu}}^{\mathrm{Fcc}}+\frac{y}{x+y}\,{}^0G_{\mathrm{Er}}^{\mathrm{Hcp}}+A+BT \quad (11)$$

where the $L_{\mathrm{Al,Cu:Er}}$ represents the interaction between Al and Cu in the first sublattice, expressed in Redlich–Kister polynomials.

4. Results and discussion

4.1. The Cu–Er binary system

On the basis of lattice stabilities cited from Dinsdale [23], the optimization of the Cu–Er system was carried out using the Parrot module in the Thermo_Calc program developed by Sundman et al. [24]. Also, we had checked our database using Pandat software [25]. The phase diagram and thermochemical data were used as input to the program for the optimization. Each piece of selected information was given a certain weight by personal judgment, and verified by a trial and error method during the assessment, until most of the selected experimental information was reproduced within the selected uncertainty limits. All the parameters evaluated are listed in Table 2.

Fig. 4 shows the calculated phase diagram of the Cu–Er system compared with experimental data, while Table 3 lists the invariant reactions in the Cu–Er system. An agreement within 9 K between the calculated and experimentally determined temperatures for these invariant reactions has been obtained. As shown in Fig. 5, the calculated mixing enthalpy in the Cu-rich region is in agreement with the experiment data. The minimum values of mixing enthalpies of liquid in the Cu–RE system (such as Cu–Dy [26] and Cu–Yb systems [27]) are about 40 at. RE% and our work can fit this tendency. It is demonstrated that the experimental data for thermodynamically properties can be well described using the present thermodynamic parameters within the experimental errors.

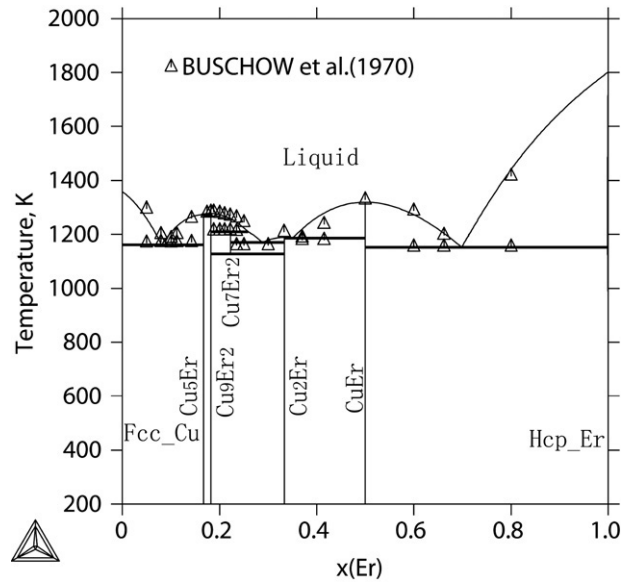

Fig. 4. The calculated Cu–Er phase diagram with experimental data [17].

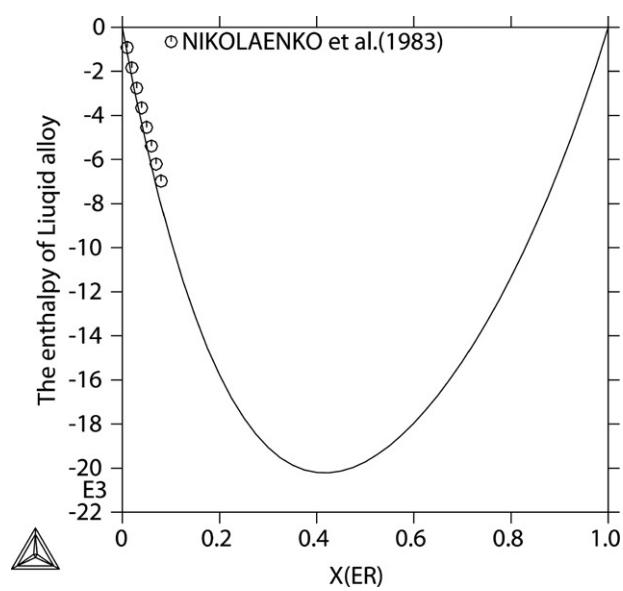

Fig. 5. The calculated mixing enthalpy of Cu–Er liquid alloy at 1473 K with experimental data [19].

4.2. The Al–Cu–Er ternary system

There is no thermochemical and liquid projection information, so the ternary interaction parameters of liquid are considered as zero. On the basis of the phase relations of Kuz'ma et al. [21], seven ternary compounds are assessed. The parameters are listed in Table 2. The calculated Al–Cu–Er isothermal section at 600 °C, with Pandat software, is shown in Fig. 6. From this we find that the homogeneity regions of Al_2Er, Cu_2Er and CuEr extend, respectively, up to 12 at.% Cu, 3.9 at.% Al and 29 at.% Al, which are in good agreement with the work of Kuz'ma et al. [21]. Two semi-stoichiometric phases, τ_2-$(Al, Cu)_{17}Er_2$ and τ_3-$(Al, Cu)_5Er$, were determined to contain about 37.6–50 at.% and 15–45 at.% Al, respectively. However, from our calculation, the homogeneity regions of these two phases are 39.2–47.8 at.% and 20.1–51.9 at.% Al. These discrepancies between experimental and calculated equilibrium compositions of ternary phases may be due to an insufficient annealing of samples.

Table 2
Thermodynamic parameters of the Al–Cu–Er system

Phase	Thermodynamic parameters	Ref.
Liquid model: (Al, Cu, Er)	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Al,Cu}} = -67\,094 + 8.555T$	[17]
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Al,Cu}} = +32\,148 - 7.118T$	[17]
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Al,Cu}} = +5915 - 5.889T$	[17]
	${}^{3}L^{\mathrm{Liq}}_{\mathrm{Al,Cu}} = -8175 + 6.049T$	[17]
	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Al,Er}} = -176\,486 + 55.6852T$	[19]
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Al,Er}} = -36\,685.5 + 23.4492T$	[19]
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Al,Er}} = +34\,349.1 - 8.23519T$	[19]
	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Cu,Er}} = -78\,882.6324 + 2.842T$	This work
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Cu,Er}} = -23\,265.2445 + 0.15T$	This work
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Cu,Er}} = -16\,190.5543 + 7.698T$	This work
Bcc model: (Al, Cu, Er)	${}^{0}L^{\mathrm{Bcc}}_{\mathrm{Al,Cu}} = -73\,554 + 4T$	[16]
	${}^{1}L^{\mathrm{Bcc}}_{\mathrm{Al,Cu}} = +51\,500 - 11.84T$	[16]
Fcc model: (Al, Cu, Er)	${}^{0}L^{\mathrm{Fcc}}_{\mathrm{Al,Cu}} = -53\,520 + 2T$	[16]
	${}^{1}L^{\mathrm{Fcc}}_{\mathrm{Al,Cu}} = +38\,590 - 2T$	[16]
	${}^{2}L^{\mathrm{Fcc}}_{\mathrm{Al,Cu}} = 1170$	[16]
$L1_2$ model: $(\mathrm{Al, Cu, Er})_{0.75}(\mathrm{Al, Cu, Er})_{0.25}$	$G^{L12}_{\mathrm{Al:Al}} = 0$	[19]
	$G^{L12}_{\mathrm{Er:Er}} = 0$	[19]
	$G^{L12}_{\mathrm{Cu:Cu}} = 0$	This work
	$G^{L12}_{\mathrm{Al:Er}} = -75\,258 + 18.4941T$	[19]
	$G^{L12}_{\mathrm{Er:Al}} = 0$	[19]
	${}^{0}L^{L12}_{\mathrm{Al:Al,Er}} = 0$	[19]
	${}^{1}L^{L12}_{\mathrm{Al:Al,Er}} = 0$	[19]
	${}^{0}L^{L12}_{\mathrm{Al,Er:Al}} = -112\,887 + 27.74115T$	[19]
	${}^{1}L^{L12}_{\mathrm{Al,Er:Al}} = -112\,887 + 27.74115T$	[19]
	${}^{0}L^{L12}_{\mathrm{Er:Al,Er}} = 0$	[19]
	${}^{1}L^{L12}_{\mathrm{Er:Al,Er}} = 0$	[19]
	${}^{0}L^{L12}_{\mathrm{Al,Er:Er}} = +112\,887 - 27.74115T$	[19]
	${}^{1}L^{L12}_{\mathrm{Al,Er:Er}} = +37\,629 - 9.24705T$	[19]
Hcp model: (Cu,Er)	${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Cu,Er}} = +15\,000$	This work
θ model: $(\mathrm{Al})_2(\mathrm{AL, Cu})_1$	$G^{\theta}_{\mathrm{Al:Al}} = +30\,249 - 14.439T + 3GHSERAL$	[16]
	$G^{\theta}_{\mathrm{Al:Cu}} = -47\,406 + 6.75T + 2GHSERAL + GHSERCU$	[16]
	${}^{0}L^{\theta}_{\mathrm{Al:Al,Cu}} = 2211$	[16]
η model: $(\mathrm{AL, Cu})_1(\mathrm{Cu})_1$	$G^{\eta}_{\mathrm{Al:Cu}} = -40\,560 + 3.14T + GHSERAL + GHSERCU$	[16]
	$G^{\eta}_{\mathrm{Cu:Cu}} = +8034 - 2.51T + 2GHSERCU$	[16]
	${}^{0}L^{\eta}_{\mathrm{Al,Cu:Cu}} = -25\,740 - 20T$	[16]
ε model: $(\mathrm{Al, Cu})_1(\mathrm{Cu})_1$	$G^{\varepsilon}_{\mathrm{Al:Cu}} = -36\,976 + 1.2T + GHSERAL + GHSERCU$	[16]
	$G^{\varepsilon}_{\mathrm{Cu:Cu}} = +8034 - 2.51T + 2GHSERCU$	[16]
	${}^{0}L^{\varepsilon}_{\mathrm{Al,Cu:Cu}} = +7600 - 24T$	[16]
	${}^{1}L^{\varepsilon}_{\mathrm{Al,Cu:Cu}} = -72\,000$	[16]
ζ model: $(\mathrm{Al})_9(\mathrm{Cu})_{11}$	$G^{\zeta}_{\mathrm{Al:Cu}} = -420\,000 + 18T + 9GHSERAL + 11GHSERCU$	[16]
δ model: $(\mathrm{Al})_2(\mathrm{Cu})_3$	$G^{\zeta}_{\mathrm{Al:Cu}} = -106\,700 + 3T + 2GHSERAL + 3GHSERCU$	[16]
γD83 model: $(\mathrm{Al})_4(\mathrm{Al, Cu})_1(\mathrm{Cu})_8$	$G^{\gamma D83}_{\mathrm{Al:Al:Cu}} = -277\,739 + 215T - 30T\ln(T) + 5GHSERAL + 8GHSERCU$	[16]
	$G^{\gamma D83}_{\mathrm{Al:Cu:Cu}} = -280\,501 + 379.6T - 52T\ln(T) + 4GHSERAL + 9GHSERCU$	[16]
γ model: $(\mathrm{Al})_4(\mathrm{Al, Cu})_1(\mathrm{Cu})_8$	$G^{\gamma}_{\mathrm{Al:Al:Cu}} = -219\,258 - 45.5T + 5GHSERAL + 8GHSERCU$	[16]
	$G^{\gamma}_{\mathrm{Al:Cu:Cu}} = -200\,460 - 58.5T + 4GHSERAL + 9GHSERCU$	[16]
$\mathrm{Al_2Er}$ model: $(\mathrm{Al, Cu, Er})_2(\mathrm{Al, Cu, Er})_1$	$G^{\mathrm{Al_2Er}}_{\mathrm{Al:Al}} = 15\,000 + 3GHSERAL$	[19]
	$G^{\mathrm{Al_2Er}}_{\mathrm{Al:Er}} = -165\,000 + 40.338T + 2GHSERAL + GHSERER$	[19]
	$G^{\mathrm{Al_2Er}}_{\mathrm{Er:Al}} = +195\,000 - 40.338T + GHSERAL + 2GHSERER$	[19]
	$G^{\mathrm{Al_2Er}}_{\mathrm{Er:Er}} = 15\,000 + 3GHSERER$	[19]
	$G^{\mathrm{Al_2Er}}_{\mathrm{Al:Cu}} = 15\,000 + 2GHSERAL + GHSERCU$	This work
	$G^{\mathrm{Al_2Er}}_{\mathrm{Cu:Al}} = 15\,000 + GHSERAL + 2GHSERCU$	This work

(*continued on next page*)

Table 2 (*continued*)

Phase	Thermodynamic parameters	Ref.
	$G^{Al_2Er}_{Cu:Cu} = 15\,000 + 3GHSERCU$	This work
	$G^{Al_2Er}_{Cu:Er} = -45\,000 + 1.1T + 2GHSERCU + GHSERER$	This work
	$G^{Al_2Er}_{Er:Cu} = -30\,000 + 0.03T + GHSERCU + 2GHSERER$	This work
	${}^0L^{Al_2Er}_{Al,Cu:Er} = -65\,500 + 2.338T$	This work
	${}^0L^{Al_2Er}_{Er:Al,Cu} = 65\,500 - 2.338T$	This work
AlEr model: $(Al)_{0.5}(Er)_{0.5}$	$G^{AlEr}_{Al:Er} = -50\,000 + 11.992178T + 0.5GHSERAL + 0.5GHSERER$	[19]
Al_2Er_3 model: $(Al)_{0.4}(Er)_{0.6}$	$G^{Al_2Er_3}_{Al:Er} = -45000 + 10.8726018T + 0.4GHSERAL + 0.6GHSERER$	[19]
$AlEr_2$ model: $(Al)_{0.333}(Er)_{0.667}$	$G^{AlEr_2}_{Al:Er} = -39\,000 + 9.2227795T + 0.333GHSERAL + 0.667GHSERER$	[19]
Cu_5Er model: $(Cu)_{0.8333}(Er)_{0.1667}$	$G^{Cu_5Er}_{Cu:Er} = -16\,392.8291 + 0.254T + 0.8333GHSERCU + 0.1667GHSERER$	This work
Cu_9Er_2 model: $(Cu)_{0.8182}(Er)_{0.1818}$	$G^{Cu_9Er_2}_{Cu:Er} = -16\,793.3827 - 0.223T + 0.8182GHSERCU + 0.1818GHSERER$	This work
Cu_7Er_2 model: $(Cu)_{0.7778}(Er)_{0.2222}$	$G^{Cu_7Er_2}_{Cu:Er} = -14\,299.99 - 3.5T + 0.7778GHSERCU + 0.2222GHSERER$	This work
Cu_2Er model: $(Al, Cu)_{0.6667}(Er)_{0.3333}$	$G^{Cu_2Er}_{Cu:Er} = -21\,343.498 + 0.01T + 0.6667GHSERCU + 0.3333GHSERER$	This work
	$G^{Cu_2Er}_{Al:Er} = -45\,000 + 12T + 0.6667GHSERAL + 0.3333GHSERER$	This work
	${}^0L^{Cu_2Er}_{Al,Cu:Er} = -13\,500$	This work
CuEr model: $(Al, Cu)_{0.5}(Er)_{0.5}$	$G^{CuEr}_{Al:Er} = -45\,000 + 11.992178T + 0.5GHSERAl + 0.5GHSERER$	This work
	$G^{CuEr}_{Cu:Er} = -26\,102.3212 + 2T + 0.5GHSERCU + 0.5GHSERER$	This work
	${}^0L^{CuEr}_{Al,Cu:Er} = -18\,090$	This work
τ_1 model: $(Al)_{0.615385}(Cu)_{0.307692}(Er)_{0.076923}$	$G^{\tau 1}_{Al:Cu:Er} = -32\,000 + 2T + 0.615385GHSERAl + 0.307692GHSERCU + 0.076923GHSERER$	This work
τ_2 model: $(Al, Cu)_{0.894737}(Er)_{0.105263}$	$G^{\tau 2}_{:Cu:Er} = -3400 + 0.7T + 0.894737GHSERCU + 0.105263GHSERER$	This work
	$G^{\tau 2}_{Al:Er} = -11\,250 + 1.7T + 0.894737GHSERAl + 0.105263GHSERER$	This work
	${}^0L^{\tau 2}_{Al,Cu:Er} = -90\,100 + 2T$	This work
τ_3 model: $(Al, Cu)_{0.833333}(Er)_{0.166667}$	$G^{\tau 3}_{Cu:Er} = -8800 + 1.2T + 0.833333GHSERCU + 0.166667GHSERER$	This work
	$G^{\tau 3}_{Al:Er} = -20\,410 + 1.7T + 0.833333GHSERAl + 0.166667GHSERER$	This work
	${}^0L^{\tau 3}_{Al,Cu:Er} = -86\,950 + 3.1T$	This work
τ_4 model: $(Al)_{0.6}(Cu)_{0.2}(Er)_{0.2}$	$G^{\tau 4}_{Al:Cu:Er} = -40\,500 + 0.81T + 0.6GHSERAl + 0.2GHSERCU + 0.2GHSERER$	This work
τ_5 model: $(Al)_{0.5}(Cu)_{0.3}(Er)_{0.2}$	$G^{\tau 5}_{Al:Cu:Er} = -42\,200 + 1.01T + 0.5GHSERAl + 0.3GHSERCU + 0.2GHSERER$	This work
τ_6 model: $(Al)_{0.45}(Cu)_{0.3}(Er)_{0.25}$	$G^{\tau 6}_{Al:Cu:Er} = -43\,500 + 1.01T + 0.45GHSERAl + 0.3GHSERCU + 0.25GHSERER$	This work
τ_7 model: $(Al)_{0.333333}(Cu)_{0.333333}(Er)_{0.333334}$	$G^{\tau 7}_{Al:Cu:Er} = -41\,000 + 0.05T + 0.333333GHSERAl + 0.333333GHSERCU + 0.333334GHSERER$	This work

Table 3
Invariant reactions in the Cu–Er system

Reaction	Composition of liquid (at.% Er)	T/K	Type	Ref.
L ⇔ Fcc_Cu + Cu_5Er	9.5	1168	Eutectic	[12]
	8.15	1162	Eutectic	This work
L + Cu_9Er_2 ⇔ Cu_5Er	16.2	1278	Peritectic	[12]
	16.4	1273	Peritectic	This work
L ⇔ Cu_9Er_2	–	1283	Congruent	[12]
	–	1278	Congruent	This work
L + Cu_9Er_2 ⇔ Cu_7Er_2	–	1213	Peritectic	[12]
	25.9	1210	Peritectic	This work
L ⇔ Cu_7Er_2 + Cu_2Er	30.	1158	Eutectic	[12]
	29.3	1160	Eutectic	This work
L ⇔ Cu_2Er + CuEr	38.5	1178	Eutectic	[12]
	34.4	1173	Eutectic	This work
L ⇔ Cu_9Er_2	–	1328	Congruent	[12]
	–	1319	Congruent	This work
L ⇔ CuEr + Hcp_Er	69.5	1153	Eutectic	[12]
	69.9	1152	Eutectic	This work

Additionally, the liquidus projection is further calculated as shown in Figs. 7 and 8. Invariant equilibria involving the Al–Cu binary phases are not visible in Fig. 7 because they are very close to the Al–Cu binary subsystem ones. And the calculated invariant reactions and temperatures involving liquid in the Al–Cu–Er ternary system are summarized in Table 4. Further experimental data are needed to verify the liquidus projection.

5. Conclusions

The Al–Cu–Er ternary system was assessed thermodynamically on the basis of reported experimental data for phase diagrams and thermodynamic properties. Reasonable agreement between calculated and experimental data was obtained and thermodynamic parameters for various phases in this ternary system were obtained.

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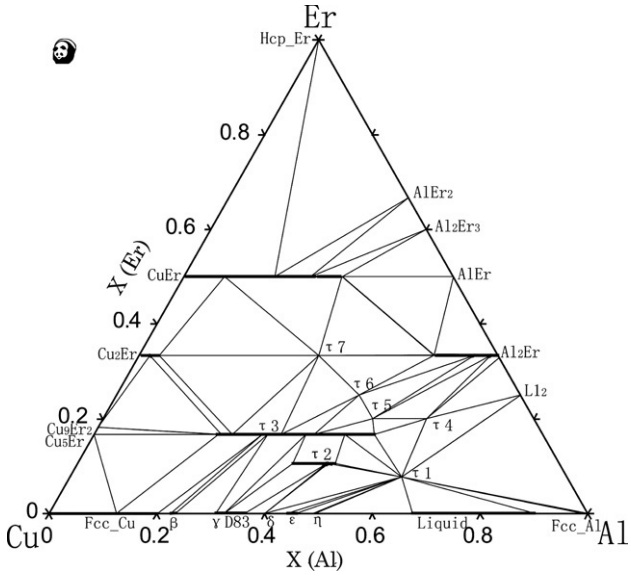

Fig. 6. The calculated isotherm section of the Al–Cu–Er system at 873 K.

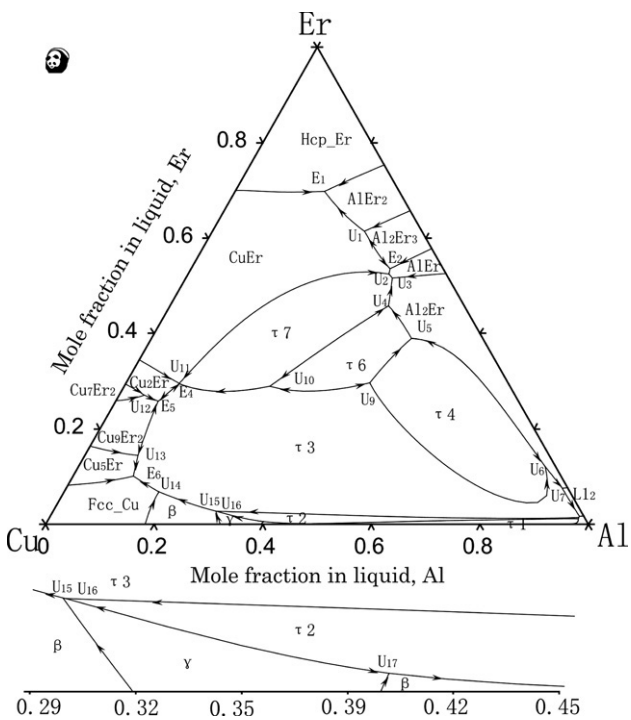

Fig. 7. The calculated liquidus projection of the Al–Cu–Er system.

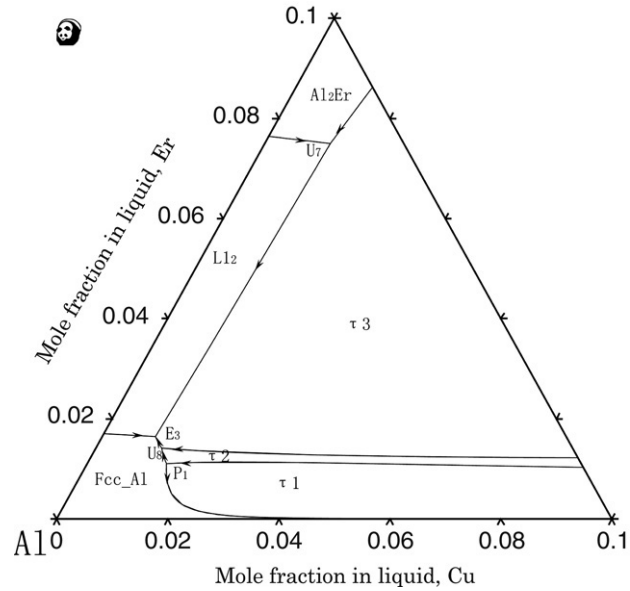

Fig. 8. The calculated liquidus projection of the Al-rich corner.

Table 4
Calculated invariant reactions and temperature of the Al–Cu–Er ternary system

Type	Reaction	T/°C
E_1	Liquid ⇔ CuEr + $AlEr_2$ + Hcp_Er	883.15
E_2	Liquid ⇔ CuEr + Al_2Er_3 + AlEr	1004.80
E_3	Liquid ⇔ $L1_2$ + τ_3 + Fcc_Al	642.24
E_4	Liquid ⇔ Cu_2Er + τ_3 + τ_7	815.50
E_5	Liquid ⇔ Cu_2Er + τ_3 + Cu_9Er_2	805.23
E_6	Liquid ⇔ Fcc_Cu + Cu_5Er + τ_3	811.52
U_1	Liquid + Al_2Er_3 ⇔ CuEr + $AlEr_2$	973.81
U_2	Liquid + τ_7 ⇔ AlEr + CuEr	1009.00
U_3	Liquid + Al_2Er ⇔ AlEr + τ_7	1023.19
U_4	Liquid + τ_6 ⇔ τ_7 + Al_2Er	1141.87
U_5	Liquid + τ_4 ⇔ Al_2Er + τ_6	1288.08
U_6	Liquid + τ_4 ⇔ τ_3 + Al_2Er	1132.56
U_7	Liquid + Al_2Er ⇔ τ_3 + $L1_2$	954.39
U_8	Liquid + τ_2 ⇔ τ_3 + $L1_2$	642.67
U_9	Liquid + τ_3 ⇔ τ_6 + τ_4	1462.87
U_{10}	Liquid + τ_6 ⇔ τ_3 + τ_7	1264.47
U_{11}	Liquid + CuEr ⇔ Cu_2Er + τ_7	815.76
U_{12}	Liquid + Cu_7Er_2 ⇔ Cu_9Er_2 + Cu_2Er	843.30
U_{13}	Liquid + Cu_9Er_2 ⇔ τ_3 + Cu_5Er	859.58
U_{14}	Liquid + β ⇔ τ_3 + Fcc_Cu	889.40
U_{15}	Liquid + γ ⇔ β + τ_3	976.56
U_{16}	Liquid + τ_2 ⇔ γ + τ_3	976.98
U_{17}	Liquid + γ ⇔ β + τ_2	946.41
U_{18}	Liquid + β ⇔ τ_2 + ε	850.90
U_{19}	Liquid + τ_2 ⇔ τ_1 + ε	850.22
U_{20}	Liquid + τ_1 ⇔ θ + Fcc_Al	547.46
P_1	Liquid + τ_1 + τ_2 ⇔ Fcc_Al	643.63
P_2	Liquid + ε + τ_1 ⇔ η	624.96
P_3	Liquid + η + τ_1 ⇔ θ	595.70

Partial calculations were performed using the Pandat program licensed from CompuThermo, LLC, Madison, WI, USA.

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